

Wind Tunnel Testing of Static Aerodynamic and Power Consumption Characteristics of an Octocopter

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The introduction of multirotor unmanned aircraft systems (UAS) in low-altitude airspace poses significant safety challenges. Further development of advanced aviation systems is needed to realize the full potential of UAS applications, particularly for urban and beyond visual line-of-sight (BVLOS) operations. The NASA System Wide Safety (SWS) project is developing key technologies to mitigate future airspace safety risks involving autonomous aircraft operation in densely populated areas. An important part of these technologies requires flight dynamics and performance models to assess feasible trajectories, power consumption requirements, and operational envelope limitations. To support these efforts, this research seeks to improve the accuracy of multirotor aerodynamic models and investigate stability characteristics relevant to flight safety. This paper presents a recent study of an octocopter vehicle using the NASA Langley 12-Foot Low-Speed Tunnel (LST). Results from isolated-airframe (no propellers) and powered-airframe tests are presented. A one-factor-at-a-time (OFAT) approach was used to explore the effect of propeller advance ratio and vehicle angle of attack on vehicle forces, moments, and power consumption. The results show nonlinear aerodynamic behavior and indicate regions where aerodynamic interaction effects may be significant.

I. Nomenclature

A	=	propeller disk area, ft ²
b	=	airframe span, ft
C_T, C_P	=	propeller thrust and power coefficients
C_{T_h}, C_{P_h}	=	hover propeller thrust and power coefficients
C_x, C_z, C_m	=	powered-airframe longitudinal body-axis aerodynamic force and moment coefficients
$C_{x_A}, C_{y_A}, C_{z_A}$	=	isolated-airframe body-axis aerodynamic force coefficients
$C_{l_A}, C_{m_A}, C_{n_A}$	=	isolated-airframe body-axis aerodynamic moment coefficients
D	=	propeller diameter, ft
J	=	propeller advance ratio
l	=	airframe length, ft
L_A, M_A, N_A	=	isolated-airframe body-axis aerodynamic moments, ft · lb
m	=	mass, slug
M	=	powered-airframe body-axis aerodynamic pitching moment, ft · lb
n	=	motor speed, revolutions per second
N_m	=	number of motors
P	=	power, W or ft · lb/s
$\bar{q}$	=	dynamic pressure, lb/ft ²
R	=	propeller radius, ft
S	=	airframe reference area, ft ²
T	=	thrust, lb
V	=	true airspeed, ft/s
X, Z	=	powered-airframe longitudinal body-axis aerodynamic forces, lb
X_A, Y_A, Z_A	=	isolated-airframe body-axis aerodynamic forces, lb
α	=	angle of attack, rad or deg

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β	= angle of sideslip, rad or deg
ρ	= air density, slug/ft ³
Ω	= motor speed, radians per second
$\bar{\Omega}$	= mean motor speed, radians per second

II. Introduction

A. Motivation

The safe integration of unmanned aircraft systems (UAS) in the national airspace system will require advanced aviation technologies. The NASA System Wide Safety (SWS) project is developing highly assured capabilities to mitigate risk in future airspace operations, including unpiloted aircraft in densely populated areas. Within this project, a collection of Services, Functions, and Capabilities (SFCs) is in development to address precursors, anomalies, or trends observed during operations [1,2]. Data from ground stations, onboard sensors, UAS traffic management (UTM) systems, external weather services, battery health monitoring systems, and aerodynamic models supports pre-flight and in-flight applications for trajectory planning, real-time risk assessment, and safety metric monitoring [3-6]. An important data requirement for flight planning is the prediction of vehicle trajectories and energy consumption in operational and out-of-envelope conditions. UAS operators and onboard decision-making functions will benefit from energy reserve predictions in path planning, rerouting, and obstacle avoidance scenarios. Trajectory predictions are useful to assess feasible tracking accuracy in the presence of atmospheric effects, subsystem failures, or control system anomalies. This is especially important for air traffic management systems that define UAS separation standards. These capabilities require aerodynamic models that account for the influence of relative wind angles, motor speeds, and complex interaction effects. Currently, there are few high-fidelity multirotor UAS flight dynamics and performance models available to meet these requirements, particularly to support analysis of electric motor power consumption.

The wide range of conditions experienced within the operational flight envelope (OFE) of multirotor UAS present challenges for aerodynamic model development. Relative wind angles and airspeeds significantly vary between hover, forward flight, vertical climb, and descent phases. Small-scale propellers have nonlinear performance characteristics that are not adequately modeled by hover performance models. In forward flight, local edgewise and axial velocity components cause asymmetric induced velocities across the propeller disk that are difficult to predict using analytical models for real-time simulation. Small UAS (sUAS) propellers operate in a low Reynolds number regime making extrapolation from full-scale rotorcraft modeling tools inaccurate. In steep descents, multirotors can enter flight conditions resulting in large unsteady thrust fluctuations and controllability issues potentially leading to vortex ring state (VRS). Additional modeling complexities arise from UAS configurations that have multiple propellers located near each other and airframe surfaces. This results in complex aerodynamic interactions that can influence important performance parameters such as energy consumption, thrust requirements, and control margins. This can also result in abrupt nonlinearities that are important to capture for flight dynamics and performance modeling applications. This research seeks to increase the accuracy of multirotor aerodynamic and power consumption models and investigate their unique stability and control characteristics.

B. Objectives and Approach

Multirotor aerodynamic modeling methods and requirements for flight dynamics and performance analysis are a current topic of NASA research. Recently, NASA wind tunnel facilities have supported isolated propeller, powered-airframe, and free-flight experiments of UAS vehicles [7,8]. One major outcome was the development of a high-fidelity quadrotor flight dynamics model integrated with a closed loop six degree-of-freedom (DOF) simulation. The simulation incorporated propulsion models that account for relative incidence angle effects and airframe contributions. Additionally, research was conducted to include aerodynamic interactions that occur between propulsors and airframe surfaces [9]. This simulation has supported studies of hazardous flight conditions for multirotor vehicles [10] and in-time risk assessment services [11]. There have been other recent experimental studies of multirotor vehicles including static wind tunnel tests of several UAS configurations [12], dynamic wind tunnel testing of a quadrotor configuration in forward flight conditions [13-15], and free-flight experiments to support system identification [16,17]. In addition to UAS vehicles, NASA research has developed aerodynamic modeling methods for vertical takeoff and landing (VTOL) vehicles for Urban Air Mobility (UAM) applications. Wind tunnel testing methods involving design of experiments (DOE), response surface methods (RSM), and system identification techniques are being advanced for aircraft with many propulsors and control effectors [18-22]. The research described in this paper aims to leverage and expand on these capabilities to improve the aerodynamic database of multirotor UAS models.

The primary objective of this work was to investigate the static aerodynamic and power consumption characteristics of a generic octocopter vehicle as a function of relative air flow angles, airspeeds, and motor speeds.

This configuration is representative of many UAS vehicles used as research platforms and for commercial applications. The approach to meet these objectives involved instrumenting the octocopter to measure forces, moments, and power consumption in the NASA Langley 12-Foot Low-Speed Tunnel (12-Foot LST). Isolated airframe (no propellers) and a powered-airframe configurations were studied using one-factor-at-a-time (OFAT) test techniques. The results from this work will be used to identify regions of nonlinear aerodynamic behavior and support future aerodynamic and power consumption model development.

This paper is organized as follows. Section III provides details of the experimental setup including the wind tunnel facility, vehicle description, model installation, and test cases. Section IV presents the results and analysis of the OFAT static wind tunnel testing. Section V summarizes key findings, conclusions, and future directions for this research.

III. Experimental Setup

A. Wind Tunnel Facility

The experiments were performed in the NASA Langley 12-Foot LST. The test section is 15 ft in length and 12 ft in height and width with an octagonal cross section. This size is suitable to study many full-scale sUAS platforms. The facility supports free-flight, static, and dynamic test techniques with a maximum operating freestream dynamic pressure of approximately 7 lb/ft². Models can mount to a rigid sting mounted on a C-strut which enables pitch orientations from approximately -88° to +88° using upright and inverted airframe orientations. The C-strut can also rotate to reach yaw angles between $\pm 90^\circ$. The facility is equipped with data acquisition systems to obtain force, moment, surface pressure, and power consumption measurements. Further details about the facility's capabilities are available online^{**}.

B. Vehicle Description

The vehicle used in this study is a commercial off-the-shelf (COTS) octocopter that is representative of commonly used multirotor UAS platforms. The airframe is a Tarot X8 model which consists of a carbon fiber center body with mounting holes for avionics, eight aluminum arms with motor mounts on each end, a battery tray, and retractable landing gear. The center body consists of two 13-inch diameter plates that are offset with 1.5-inch vertical posts. The eight foldable aluminum arms are 16 inches long with 320KV brushless DC motors mounted on each end. The carbon fiber propellers have a 15-inch diameter and 5.5-inch pitch. The retractable landing gear uses two embedded servo motors to switch between extended or retracted positions. The vehicle mass and reference dimensions are provided in Table 1. The mass (m) is based on the gross takeoff weight listed by the manufacturer. The span (b) is defined as the lateral distance between the vehicle geometric center and left/right motors (i.e., motors 3, 4, 7, and 8), and was used to define the airframe reference area ($S = \pi b^2$). The rotor radius (R) was used to define the rotor disk area ($A = \pi R^2$). The length (l) is the longitudinal distance between the geometric center and the front/rear motors (i.e., motors 1, 2, 5, and 6). The reference dimensions and motor numbers are shown on a top view schematic of the octocopter in Fig. 1.

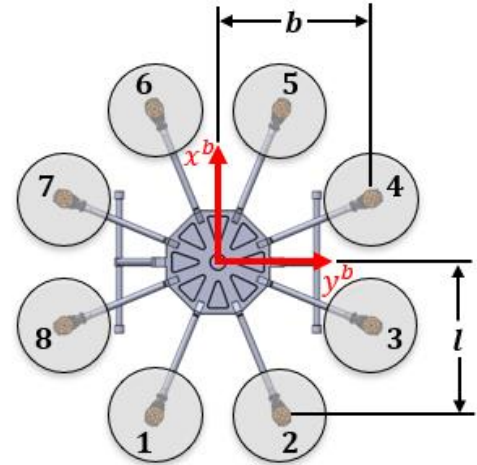


Fig. 1 Top view of schematic.

Table 1. Mass and geometric properties

m (slug)	b (ft)	l (ft)	R (ft)	S (ft ²)	A (ft ²)
0.685	1.72	1.72	0.625	9.29	1.23

^{**} Information available online at <https://researchdirectoratelarc.nasa.gov/> [retrieved 5 January 2023].

C. Model Installation and Instrumentation

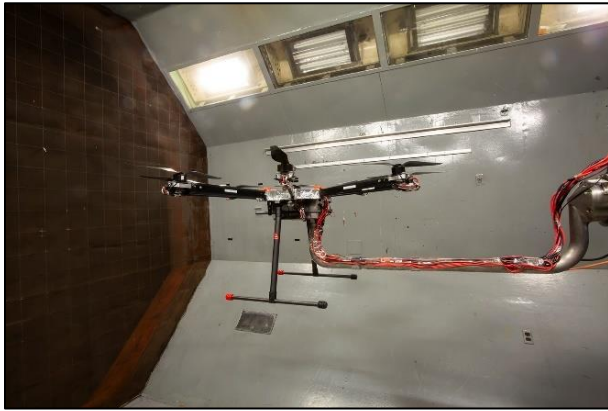
Figures 2(a) and 2(b) show side and front views of the isolated-airframe (no propellers or instrumentation) mounted in the NASA Langley 12-Foot LST test section. Figures 2(c) and 2(d) show side and rear views of the powered-airframe configuration. All testing was completed with the landing gear extended. The vehicle was mounted using a “bent” sting to minimize the influence of sting and C-strut interference effects on sensor measurements. This maximized the clearance between the sting and the propeller disk plane, resulting in a distance greater than the propeller diameter. The model was mounted to the sting with an attachment located underneath the airframe centerplate. An extension to the sting was added to increase the longitudinal distance between the model and the C-strut. During static powered-airframe testing, the rigidly mounted aircraft may experience significantly higher loads than in its designed free-flight environment due to inertial effects and structural vibrations; therefore, the airframe was modified to increase the structural stiffness. Aluminum tape was used to join the top and bottom center plates to increase torsional rigidity. An aluminum plate was fastened to the bottom carbon fiber centerplate so that a six-component strain gauge force balance and adapter could be attached underneath. The balance was selected based on expected aerodynamic and propulsive loads. The accuracy for each axis varied from 0.03% to 0.11% based on the full load range. The force and moment measurements were sampled at 1000 Hz. The balance adapter was designed to have similar dimensions and geometric location as a battery used in flight. The original motor mounts and connections to the arms were replaced with aluminum parts that were machined to the same planform dimensions.



(a) Isolated-airframe side view



(b) Isolated-airframe front view



(c) Powered-airframe side view



(d) Powered-airframe rear view

Fig 2. Octocopter model mounted in the NASA Langley 12-Foot Low-Speed Tunnel

(Credit: Lee Pollard, NASA Langley Research Center)

The motors and onboard instrumentation were powered using external 30V and 12V power supplies located outside of the test section. The power cables entered the test section near the base of the C-strut and were routed along the sting into the back of the model. The 30V power supply provided voltage and current measurements to the host computer and motor speed measurements were sampled at 90 Hz. The measurements from the strain gauge balance

and power supply were time averaged for each static point. The power measurement represents the product of voltage and current output by the power supply to the electronic speed controllers (ESC) and motors.

Motor speed control was required for the powered-airframe testing so that the propeller advance ratio could remain constant under different blade loading conditions. To support this, a proportional-integral (PI) motor speed controller with feedback measurements was implemented using onboard instrumentation. The model was instrumented with an Arduino Mega, Teensy, and eight Castle Creations Phoenix Edge 50-amp ESCs. The Teensy computed motor speeds in real-time using discrete pulses output from each ESC. Software on the Arduino Mega received the motor speed measurements from the Teensy and applied the PI control logic to generate pulse width modulation (PWM) signals.

D. Isolated-Airframe Test Conditions

The primary objective of the isolated-airframe phase of testing was to investigate the airframe aerodynamic stability characteristics. This data will also be used to support an aerodynamic database for full-envelope multirotor flight dynamic simulations. It is important to recognize that the airframe forces/moments are often significantly less than the propulsive contributions; however, their importance increases at higher airspeeds and for modeling off-nominal trajectories during power failures. The isolated-airframe was tested at angles of attack (α) ranging from -88° to $+88^\circ$ and at sideslip angles (β) between -90° to $+90^\circ$. The test section dynamic pressure ($\bar{q}$) was 4 lb/ft^2 for all test points. The onboard instrumentation, propellers, and ESCs were not mounted or installed during this phase of testing.

E. Powered-Airframe Test Conditions

The primary objective of the powered-airframe testing was to explore the force, moment, and power consumption trends as a function of vehicle angle of attack at constant propeller advance ratios (J), which is defined in Eq. (1) as a function of the freestream velocity (V), motor speed in rev/s (n) and propeller diameter (D).

$$J = V/(nD) \quad (1)$$

Table 2 lists the angle of attack range, motor speed, test section velocity, and freestream dynamic pressure for each advance ratio case. Nonlinearities observed in this data set will be used to inform future test matrix designs and regions where high data resolution may be required to capture abrupt nonlinearities. To avoid exceeding dynamic load limits on the strain gage balance, the mean motor speed across all runs was approximately 50 rev/s (or 3000 RPM) and the test section freestream dynamic pressure was limited to 4 lb/ft^2 . There were noticeably higher vibrations and acoustic effects at positive angles of attack (based on observations in the control room), potentially due to recirculation of propeller wake. Because of this, the maximum angle of attack limit was limited to 30° for all testing to avoid excessive vibration and dynamic loads. All powered-airframe testing was completed at $\beta = 0$.

Table 2 Powered-Airframe Test Conditions

J	α (deg)	n (rev/s)	V (ft/s)	$\bar{q}$ (lb/ft ²)
0.2	-88 to 30	57.7	14.2	0.25
0.3	-88 to 30	54.4	21.1	0.5
0.4	-88 to 30	57.7	28.7	1
0.5	-88 to 30	46.2	29.2	1
0.75	-88 to 30	43.5	41.1	2
1	-60 to 30	46.2	57.8	4

IV. Wind Tunnel Test Results

A. Nondimensionalization

Currently there are no widely adopted reference airframe area or dimensions to compare multirotor UAS vehicles of different sizes. This is in part due to the wide variety of configurations containing different numbers of motors, relative spacings, and airframe geometries. In this work, the isolated-airframe aerodynamic body-axis force coefficients (C_{x_A} , C_{y_A} , and C_{z_A}) and isolated-airframe aerodynamic rolling and pitching moment coefficients (C_{l_A} , C_{m_A}) were nondimensionalized using Eqs. (2)-(3).

$$C_{x_A} = \frac{X_A}{\bar{q}S}, \quad C_{y_A} = \frac{Y_A}{\bar{q}S}, \quad C_{z_A} = \frac{Z_A}{\bar{q}S} \quad (2)$$

$$C_{m_A} = \frac{M_A}{l\bar{q}S}, \quad C_{l_A} = \frac{L_A}{b\bar{q}S} \quad (3)$$

where X_A , Y_A , and Z_A are the isolated-airframe aerodynamic body-axis forces, L_A and M_A are the isolated-airframe aerodynamic body-axis moments, and the reference dimensions are defined in Table 1.

It is also useful to nondimensionalize the powered-airframe force, moment, and power measurements to compare results between different operating conditions and vehicle configurations; however, it is difficult to apply conventional propeller and lifting surface nondimensionalization approaches if individual propeller and airframe force/moment contributions are not directly measured. The presence of propeller and airframe aerodynamic interactions further complicates nondimensionalization processes because each propulsor may experience a different relative flow velocity and angle. In addition, the airframe components downstream of the propeller disk plane experience a different flow angle and dynamic pressure than the corresponding freestream values. Conventional propeller theory uses propeller thrust (C_T) and power coefficients (C_P) to compare propellers of different geometric parameters such as blade twist, chord taper, radius, and airfoil cross section [Eq. (4)].

$$C_T = \frac{T}{\rho\pi R^2(\Omega R)^2}, \quad C_P = \frac{P}{\rho\pi R^2(\Omega R)^3} \quad (4)$$

These definitions were adapted to support system identification of a quadrotor configuration using free-flight data in [16] and were also applied to the powered-airframe results presented in this paper. The dimensional response variables from the powered-airframe wind tunnel experiments were the body-axis aerodynamic forces (X and Z), body-axis aerodynamic pitching moment (M), and total electric motor power consumption (P). The powered-airframe measurements were converted to force, moment, and power coefficients using Eqs. (5)-(6). In Eq. (5), the powered-airframe force coefficients (C_x and C_z) are defined in a similar manner as C_T ; however, the denominator includes a term to represent the total number of motors (N_m) and uses the average motor speed ($\bar{\Omega}$) instead of individual speeds. The powered-airframe pitching moment coefficient (C_m) and power coefficient (C_p) are defined in Eq. (6).

$$C_x = \frac{X}{\rho N_m \pi R^2 (\bar{\Omega} R)^2}, \quad C_z = \frac{Z}{\rho N_m \pi R^2 (\bar{\Omega} R)^2} \quad (5)$$

$$C_m = \frac{M}{\rho l N_m \pi R^2 (\bar{\Omega} R)^2}, \quad C_p = \frac{P}{\rho N_m \pi R^2 (\bar{\Omega} R)^3} \quad (6)$$

B. Isolated-Airframe Results

The isolated-airframe force and moment coefficients are presented in Fig. 3. Yawing coefficient data is omitted from the figure because the data was primarily noise. The airframe data exhibits symmetry that matches with expectations based on the geometry of the airframe. The longitudinal forces and moments are symmetric about the x-axis, while the side-force and rolling moment coefficients are anti-symmetric. Furthermore, the side-force and rolling moment data at $\alpha = 0^\circ$ and $\beta = 90^\circ$ exhibit similar characteristics to the body-z force and pitching moment data at $\alpha = 0^\circ$ and $\beta = 0^\circ$, though with larger magnitude. This is likely a result of the increased cross-sectional area of the airframe due to the landing gear and battery tray at this orientation.

Figure 3(c) shows that the body-z force is decreasing with angle of attack linearly between $\alpha = -30^\circ$ and $\alpha = 30^\circ$ with a slope smaller in magnitude than an airfoil. The pitching moment coefficient data in Fig. 3(e) shows a positive slope with respect to angle of attack for $\alpha \in [-30, 30]$ deg, indicating static longitudinal instability. The airframe is also statically unstable in the roll axis, as indicated by the positive rolling moment slope with respect to sideslip for all angles of attack, as shown in Fig. 3(d).

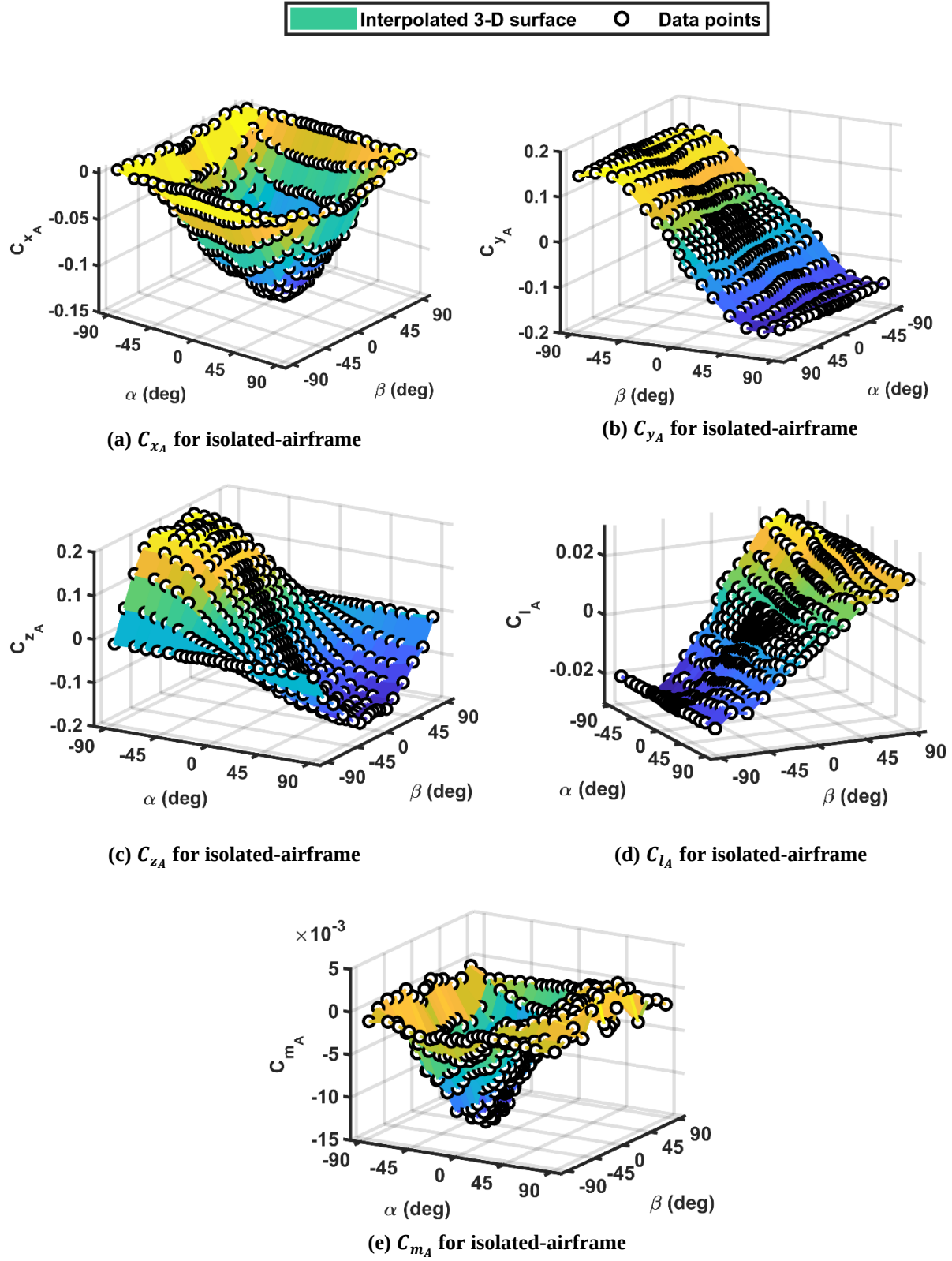


Fig 3. Isolated-airframe force and moment coefficients for $\alpha \in [-88, 88]$ deg and $\beta \in [-90, 90]$ deg. The circles represent data points.

C. Powered-Airframe Results

Figure 4 shows the measured total vehicle thrust and power consumption as a function of motor speeds for wind-off conditions ($V = 0$ ft/s). All motor speeds were uniformly increased from 0-520 rad/s. The hover thrust coefficient ($C_{T_h} = 0.0132$) and power coefficient ($C_{p_h} = 0.023$) were determined from this data using Eq. (4) as the model structure and ordinary least squares for parameter estimation. These coefficients were used to compute predicted thrust ($\hat{T}$) and power ($\hat{P}$), which show good agreement with the powered-airframe measurements.

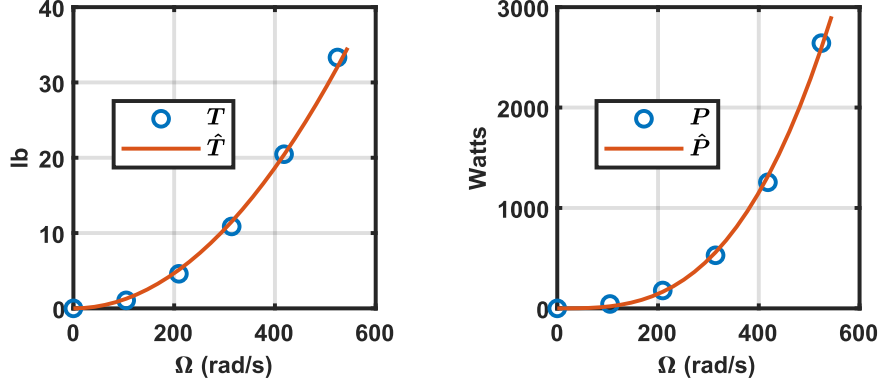


Fig. 4 Total vehicle thrust and power consumption as a function of motor speed ($V = 0$ ft/s)

The longitudinal aerodynamic forces, pitching moment, and total vehicle power consumption were time averaged for each test point and are presented in Fig. 5 in coefficient form. Figure 5(a) shows C_z versus α for each advance ratio case. The slope of C_z ($C_{z\alpha}$) is negative for this α interval, which may indicate that the propellers produce higher thrust as the relative propeller incidence angle decreases. Additionally, propeller advance ratio significantly influences the magnitude of $C_{z\alpha}$, suggesting propeller thrust may be more sensitive to variations in flow angle at higher airspeeds. Interestingly, the data points in Fig. 5(a) intercept near $\alpha = -15^\circ$ and appear to represent an inflection point. It is important to note that the aerodynamic force and moment measurements include airframe contributions that could influence $C_{z\alpha}$ at different advance ratios.

Similar trends are observed with the power coefficient, shown in in Fig. 5(b). The power coefficient increases with angle of attack, and the slope ($C_{p\alpha}$) increases with propeller advance ratio. Higher power coefficients signify an increase in motor shaft torque requirements to maintain motor speed. The power coefficient approaches zero as angle of attack approaches $\alpha = -90^\circ$ for $J = 0.75$ and $J = 1$. At these wind angles, the axial velocity component entering the propeller disk causes wind milling effects in which the free stream velocity augments the propeller rotation, resulting in a lower torque requirement. This phenomenon limited the ability to test at $\alpha < -50^\circ$ for $J = 1$ because the motor speed control system attempted to command negative throttle commands, resulting in ESC complications. The data in Fig. 5(b) indicate that increases in advances ratios beyond approximately $\alpha = -20^\circ$ result in higher power requirements, whereas the opposite trend is observed for $\alpha < -20^\circ$.

The total body-axis pitching moment is primarily caused by differential thrust between front and rear propellers relative to the vehicle center of gravity. The motor rotational speed setpoint was constant for all eight motors during the OFAT testing, which ideally results in uniform thrust production and power requirements across all propulsors; however, the presence of aerodynamic interactions in certain angle of attack regions causes changes in performance between propulsors. Figure 5(c) shows that there is a nonlinear relationship between pitching moment coefficient, angle of attack, and advance ratio. The pitching moment coefficient approaches zero as angle of attack approaches -90° , indicating the propellers are producing almost equal thrust and that propulsive interactions are not significant at this relative flow angle. The pitching moment coefficient increases at nearly a constant slope for $\alpha \in [-90^\circ, -50^\circ]$ for all values of J . The C_m contribution in this α region could primarily be local hub moments generated from asymmetric lift distribution across the propeller disk, which is commonly encountered when edgewise flow components are present. The airframe also produces a notable pitching moment contribution, likely from the landing gear.

As angle of attack increases beyond $\alpha = -50^\circ$, the effects of advance ratio and aerodynamic interactions become more apparent. The slope of C_m versus α ($C_{m\alpha}$) is positive, and increases as J increases for $\alpha \in [-50^\circ, 0^\circ]$. Wake interaction between front/rear propellers and the airframe may influence the inflow properties of the rear propellers. This could result in degraded thrust of the rear propellers and an increase in nose-up pitching moment. There is an

abrupt change in sign of $C_{m\alpha}$ that occurs at a slightly different angles of attack for each advance ratio case, ranging from approximately $\alpha = 0^\circ$ to $\alpha = 10^\circ$. The change in sign of $C_{m\alpha}$ may indicate the orientation that the rear propellers exit the influence of the front propellers, resulting in an increase in rear propeller thrust and sudden decrease in nose-up pitching moment. Flow visualization and individual propeller measurements are needed to verify these observations. Operating conditions in which propulsive interactions cause rear propeller thrust degradation are important to understand and account for in control design applications, especially since multirotor vehicles can rapidly change angle of attack when maneuvering.

Figure 5(d) presents the force coefficient in the body-axis x-direction versus angle of attack for $J \in [0.2, 1]$. The magnitude of C_x increases with advance ratio, which is due to an increase in free stream velocity relative to the tip motor speed. Additionally, increases in free stream velocity result in higher drag from the airframe and propeller blades. A minimum C_x occurs at approximately $\alpha = -15^\circ$ for each advance ratio case, which is a similar angle as the intersection point observed in Fig. 5(a). The sign of $C_{x\alpha}$ changes from negative to positive at the local minimum.

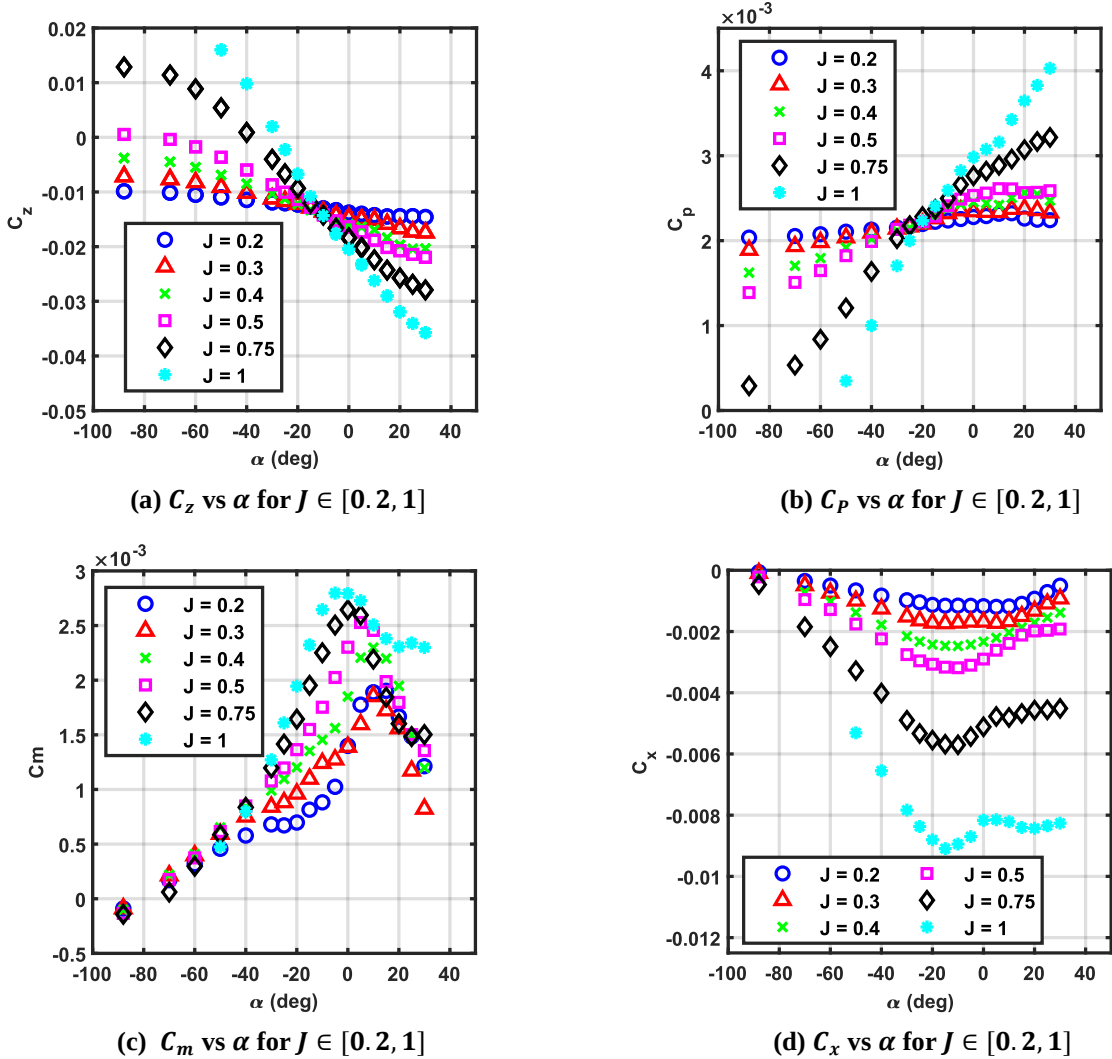


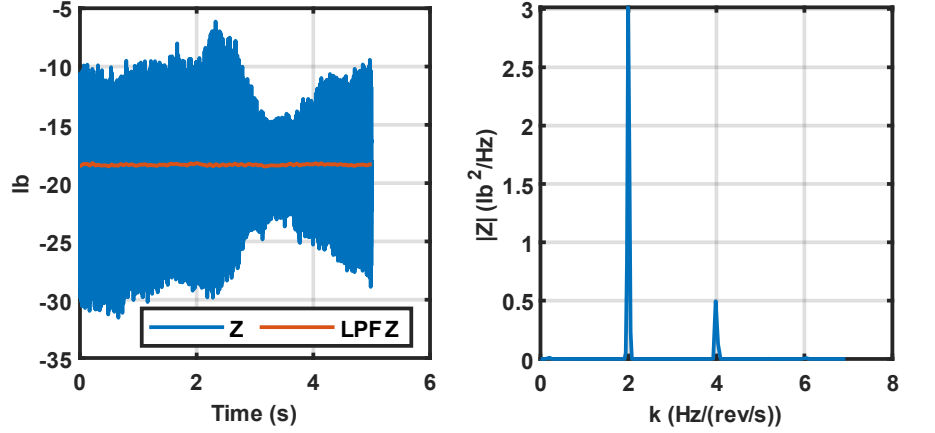
Fig. 5 Powered-airframe test results.

D. Vibration Characterization

Wind tunnel testing of the full powered-airframe resulted in large amplitude fluctuations of vehicle forces and moments. A preliminary assessment of the vibration characteristics was conducted using time series data and power

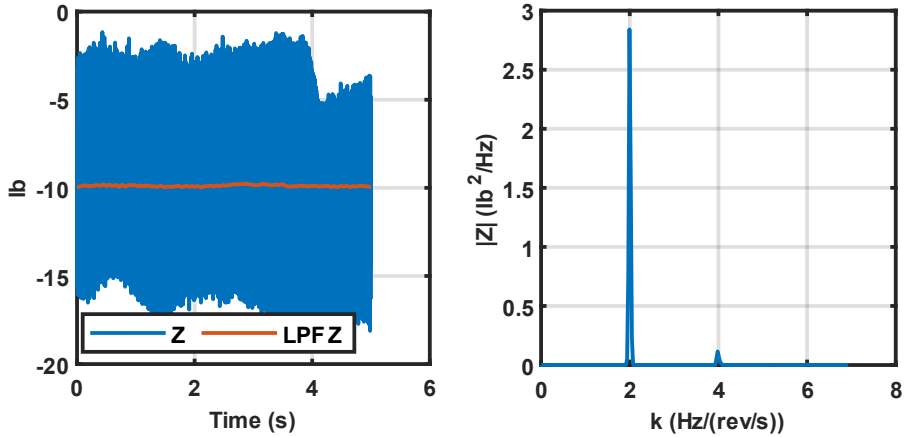
spectral density (PSD) estimates. Figures 6(a) and 7(a) show the balance measurements in the z-direction for a five second time interval for $J = 0.4$ at $\alpha = -5$ degrees and $\alpha = -30$ degrees, respectively.

The filtered data was obtained using a low-pass filter (LPF) with a cutoff frequency of 6 Hz. At $\alpha = -5$ degrees, the mean force measurement is approximately -18 lb; however, there are fluctuations that range from approximately -6 to -30 lb. Similarly, at $\alpha = -30$ degrees, the mean force measurement is -10 lb, and there are fluctuations between -2.5 to -18 lb. The filtered data stays relatively constant during the recorded five second time interval even though the amplitude of the fluctuations varies with time. Figures 6(b) and 7(b) show the PSD as a function of the motor speed harmonic ($k = f/n$), where f is the frequency vector in Hz. This shows that the primary amplitudes occur at 2x and 4x the motor speeds for that data point. Similar PSD results were found in [7]. This may be caused by the period of rotation of the two-bladed propellers. Future work will include further analysis and characterization of the vibrations at different operating conditions, particularly to explore regions where thrust fluctuations may be most dominant.



(a) Raw and filtered measurements (b) PSD versus motor speed harmonic

Fig. 6 Z time series measurements and PSD for $\alpha = -5$ deg, $J = 0.4$



(a) Raw and filtered measurements (b) PSD versus motor speed harmonic

Fig. 7 Z time series measurements and PSD for $\alpha = -30$ deg, $J = 0.4$

V. Conclusions

OFAT wind tunnel test techniques were applied to a COTS octocopter vehicle to study the isolated-airframe and powered-airframe aerodynamic characteristics. An aerodynamic database for the isolated-airframe was developed for a wide range of wind angles, which will be useful to support full-envelope flight dynamic simulations. The powered-airframe results showed variations in vehicle stability and performance that may be a result of complex aerodynamic interaction and incidence angle effects. Further research on characterizing rotor-rotor wake interactions at various separation distances and relative flow angles should be explored to better understand their influence on nonlinear

changes of forces and moments. Additionally, future research will focus on expanding the aerodynamic database to include nonzero sideslip conditions, dynamic stability characteristics, and application of DOE techniques to support aerodynamic and power consumption model development.

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